

User Guide of PCS215 PSCAD Model

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Date: Oct 21st, 2024





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- Overview
- Simulation Model Introduction
- Simulation Setting

Overview

- PSCAD model files

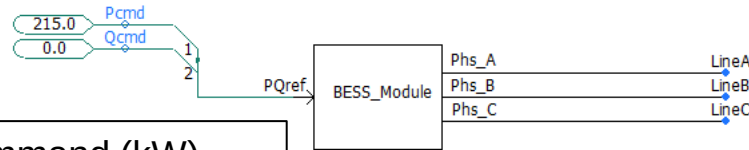
☐ 名称	修改日期	类型	大小
lib	2024/10/18 19:51	文件夹	
Duty_Limiter_f	2024/7/30 14:59	F 文件	1 KB
Mod_Ctrl_c	2024/9/13 8:46	C Source	4 KB
Mod_Ctrl_f	2024/9/13 8:40	F 文件	2 KB
PCS215_20241018	2024/10/18 19:53	PSCAD File	1,161 KB
Sys_Ctrl_c	2024/9/30 10:48	C Source	4 KB
Sys_Ctrl_f	2024/9/13 11:27	F 文件	2 KB

- Libraries in “lib” file

☐ 名称	修改日期	类型	大小
PCS_Ctrl_Mod	2024/10/18 18:02	Object File Library	910 KB
PCS_Ctrl_Sys	2024/10/18 18:24	Object File Library	678 KB

Simulation Model Introduction (I)

- PSCAD simulation model overview
- Battery + PCS PSCAD model



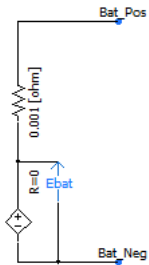
- Pcmd: active power command (kW)
- Qcmd: reactive power command (kVar)

AC side connected to grid

- Rated line-to-line voltage RMS is 690V
- Rated frequency is 60Hz

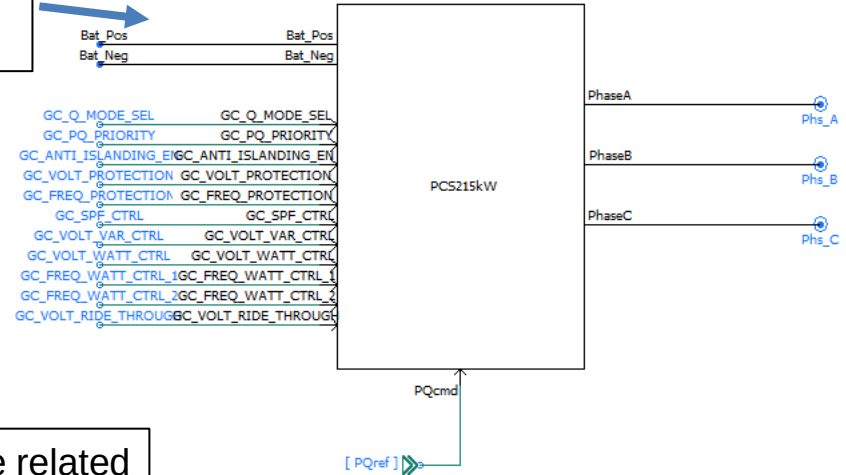
Simulation Model Introduction (I)

- PSCAD simulation model overview
- Battery PSCAD model
- PCS215kW PSCAD model



DC side connected to battery

- Rated dc voltage is 1300V



Grid code related parameters

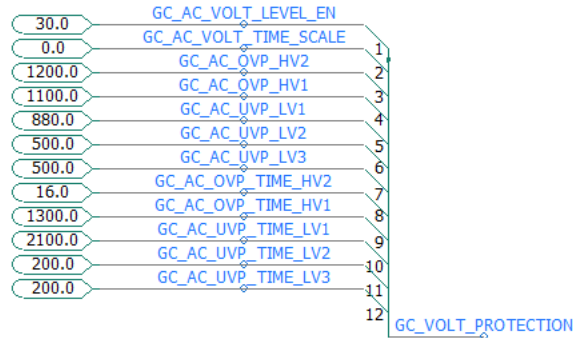
Simulation Model Introduction (II)

- PSCAD simulation model overview
- Grid code related parameters setting



Simulation Model Introduction (II)

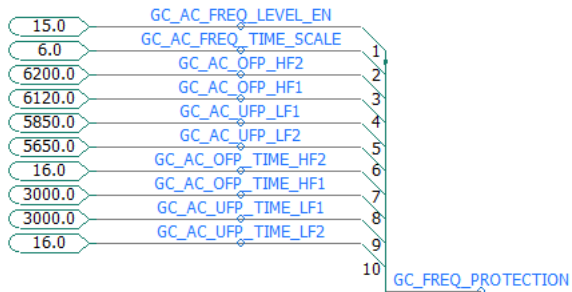
- PSCAD simulation model overview
- Grid code related parameters setting overview



ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A804	43012	Uint16	GC_AC_VOLT_LEVEL_EN	Over/Under Voltage Level Configuration OVP: 2 Levels. UVP: 3 Levels. Protection (PROT) Switch Enable bitx = 1: Check enable, bitx = 0: Check disable bit4: Over Level 2 enable bit3: Over Level 1 enable bit2: Under Level 1 enable bit1: Under Level 2 enable bit0: Under Level 3 enable	NA	RW	0	55	51
A805	43013	Uint16	GC_AC_VOLT_TIME_SCALE	Over/Under Voltage Time Scale Configuration OVP: 2 Levels. UVP: 3 Levels. Protection(PROT)/Ride Through(VRT) Time Scale bitx = 1: 0.1s(100ms), bitx = 0: 0.01s(10ms) bit4: Over Level 2 scale bit3: Over Level 1 scale bit2: Under Level 1 scale bit1: Under Level 2 scale bit0: Under Level 3 scale	NA	RW	0	55	0
A806	43014	Uint16	GC_AC_OVP_HV2	Over Voltage Protection High Value-Level 2 Configured by bit4 of Reg 43012	0.1%	RW	1200	1200	1200
A807	43015	Uint16	GC_AC_OVP_HV1	Over Voltage Protection High Value-Level 1 Configured by bit3 of Reg 43012	0.1%	RW	1100	1200	1100
A808	43016	Uint16	GC_AC_UVP_LV1	Under Voltage Protection Low Value-Level 1 Configured by bit2 of Reg 43012	0.1%	RW	50	880	880
A809	43017	Uint16	GC_AC_UVP_LV2	Under Voltage Protection Low Value-Level 2 Configured by bit1 of Reg 43012	0.1%	RW	50	500	500
A80A	43018	Uint16	GC_AC_UVP_LV3	Under Voltage Protection Low Value-Level 3 Configured by bit0 of Reg 43012	0.1%	RW	50	500	500
A80B	43019	Uint16	GC_AC_OVP_TIME_HV2	Over Voltage Protection Time High Value-Level 2 Configured by bit4 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	16	16
A80C	43020	Uint16	GC_AC_OVP_TIME_HV1	Over Voltage Protection Time High Value-Level 1 Configured by bit3 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	100	1300	1300
A80D	43021	Uint16	GC_AC_UVP_TIME_LV1	Under Voltage Protection Time Low Value-Level 1 Configured by bit2 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	200	5000	2100
A80E	43022	Uint16	GC_AC_UVP_TIME_LV2	Under Voltage Protection Time Low Value-Level 2 Configured by bit1 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	2100	200
A80F	43023	Uint16	GC_AC_UVP_TIME_LV3	Under Voltage Protection Time Low Value-Level 3 Configured by bit0 of Reg 43013 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	2100	200

Simulation Model Introduction (II)

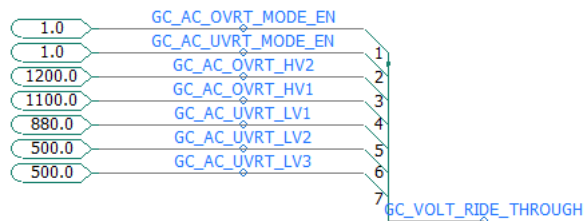
- PSCAD simulation model overview
- Grid code related parameters setting overview



ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A810	43024	Uint16	GC_AC_FREQ_LEVEL_EN	Over/Under Frequency Level Configuration OFF: 2 Levels. UFP: 2 Levels. Protection (PROT) Switch Enable bitx = 1: Check enable, bitx = 0: Check disable bit3: Over Level 2 enable bit2: Over Level 1 enable bit1: Under Level 1 enable bit0: Under Level 2 enable	NA	RW	0	51	51
A811	43025	Uint16	GC_AC_FREQ_TIME_SCALE	Over/Under Frequency Time Scale Configuration OVP: 2 Levels. UVP: 2 Levels. Protection(PROT) Ride Through(FRT) Time Scale bitx = 1: 0.1s(100ms), bitx = 0: 0.01s(10ms) bit3: Over Level 2 scale bit2: Over Level 1 scale bit1: Under Level 1 scale bit0: Under Level 2 scale	NA	RW	0	51	17
A812	43026	Uint16	GC_AC_OFF_HF2	Over Frequency Protection High Value-Level 2 Config by bit3 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	6180	6600	6200
A813	43027	Uint16	GC_AC_OFF_HF1	Over Frequency Protection High Value-Level 1 Config by bit2 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	6100	6600	6120
A814	43028	Uint16	GC_AC_UFP_LF1	Under Frequency Protection Low Value-Level 1 Config by bit1 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	5000	5900	5850
A815	43029	Uint16	GC_AC_UFP_LF2	Under Frequency Protection Low Value-Level 2 Config by bit0 of Reg 43024 Unit: 0.01Hz	0.01Hz	RW	5000	5700	5650
A816	43030	Uint16	GC_AC_OFF_TIME_HF2	Over Frequency Protection Time High Value-Level 2 Config by bit3 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	10000	16
A817	43031	Uint16	GC_AC_OFF_TIME_HF1	Over Frequency Protection Time High Value-Level 1 Config by bit2 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	1800	10000	3000
A818	43032	Uint16	GC_AC_UFP_TIME_LF1	Under Frequency Protection Time Low Value-Level 1 Config by bit1 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	1800	10000	3000
A819	43033	Uint16	GC_AC_UFP_TIME_LF2	Under Frequency Protection Time Low Value-Level 2 Config by bit0 of Reg 43025 Unit: 0.01s(10ms), 0.1s(100ms)	0.1/0.01s	RW	16	10000	16

Simulation Model Introduction (II)

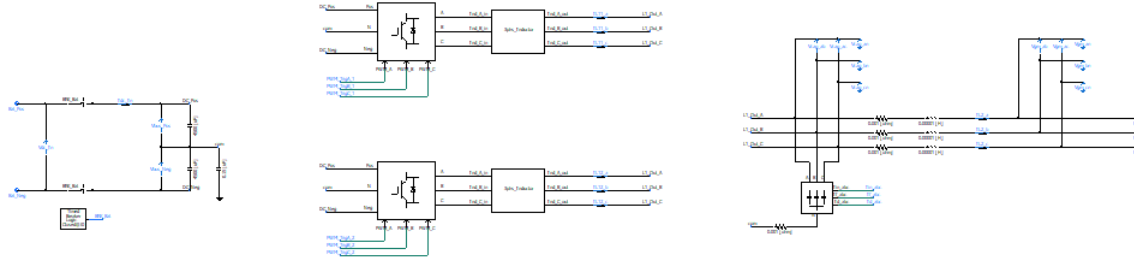
- PSCAD simulation model overview
- Grid code related parameters setting overview



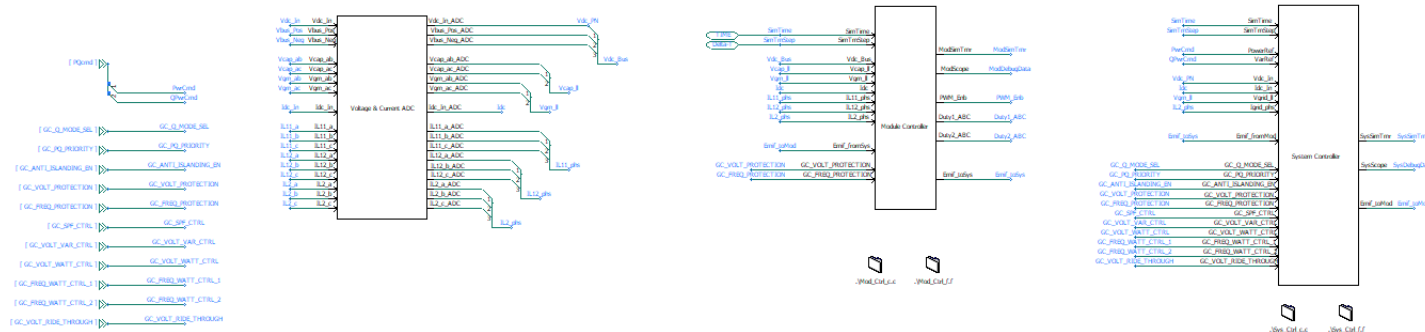
ADDR (HEX)	ADDR (DEC)	TYPE	DEFINITION	DESCRIPTION	Unit	RW	Min	Max	Default
A87B	43131	Uint16	GC_AC_OVRT_MODE_EN	Over Voltage Ride Through Mode Selection/Enable 0: Disable. 1: Enable with power limitation.	NA	RW	0	1	0
A87C	43132	Uint16	GC_AC_UVRT_MODE_EN	Under Voltage Ride Through Mode Selection/Enable. 0: Disable. 1: Enable with power limitation.	NA	RW	0	1	0
A87D	43133	Uint16	GC_AC_OVRT_HV2	OVRT Level 2 Voltage Setting	0.1%	RW	1100	1300	1200
A87E	43134	Uint16	GC_AC_OVRT_HV1	OVRT Level 1 Voltage Setting	0.1%	RW	1000	1100	1100
A87F	43135	Uint16	GC_AC_UVRT_LV1	UVRT Level 1 Voltage Setting	0.1%	RW	700	1000	880
A880	43136	Uint16	GC_AC_UVRT_LV2	UVRT Level 2 Voltage Setting	0.1%	RW	0	880	500
A881	43137	Uint16	GC_AC_UVRT_LV3	UVRT Level 3 Voltage Setting	0.1%	RW	0	500	500

Simulation Model Introduction (III)

- PCS215kW detailed simulation model



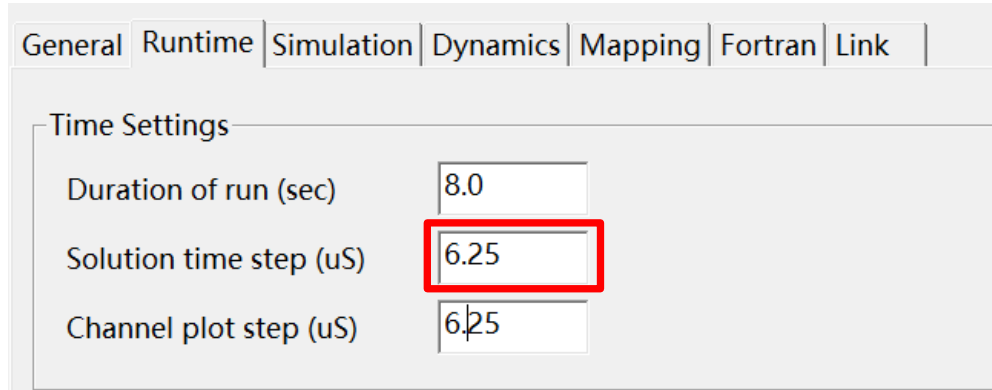
Power circuit



Controller

Simulation Setting

- Solution time step setting



The screenshot shows a software interface with a tabbed menu at the top: General, Runtime, Simulation, Dynamics, Mapping, Fortran, and Link. The 'Simulation' tab is selected. Below the tabs is a section titled 'Time Settings' containing three input fields:

Parameter	Value
Duration of run (sec)	8.0
Solution time step (uS)	6.25
Channel plot step (uS)	6.25

The 'Solution time step (uS)' field is highlighted with a red rectangular border.

- Solution time step should be **1.25us**, **1.5625us**, **3.125us** or **6.25us**
- Solution time step should be no larger than **6.25us** to guarantee the simulation accuracy

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English



Traditional
Chinese



Simplified
Chinese

